

CHAPTER 2: WEBSITE DEVELOPMENT PROCESS

Group A: Short Questions (2 Marks Each)

1. Define the term “Sitemap” in the planning phase.

A sitemap is a visual or hierarchical outline of a website’s pages and their relationships. It helps plan navigation structure and content organization before development begins.

2. What is the purpose of the “Launch” phase in web development?

The launch phase involves deploying the final website to a live server, making it publicly accessible. It includes final checks, DNS configuration, performance optimization, and announcing the site’s availability.

3. Why is “Documentation” important in the development lifecycle?

Documentation records design decisions, code structure, user guides, and technical specs. It aids future maintenance, onboarding of new developers, troubleshooting, and ensures continuity if team members change.

4. Differentiate between Front-end and Back-end development.

- **Front-end:** Focuses on the user interface (UI) and user experience (UX)—what users see and interact with (HTML, CSS, JavaScript).
 - **Back-end:** Handles server-side logic, databases, and application functionality (e.g., user authentication, data processing) using languages like PHP, Python, or Node.js.
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Group B: Long Questions (4 Marks Each)

5. “Planning is the most critical phase of web development.” Discuss this statement by listing the key activities performed during the planning phase.

Planning prevents costly mistakes later. Key activities include:

- Defining project goals and scope
- Identifying target audience
- Conducting competitor analysis
- Creating sitemaps and wireframes
- Choosing technologies and tools
- Estimating budget and timeline

Without proper planning, the site may miss user needs, exceed budget, or require major rework.

6. Explain the iterative nature of the Maintenance phase. Why is a website never truly “finished”?

Websites require ongoing updates for security patches, content refreshes, bug fixes, compatibility with new browsers/devices, and evolving user expectations. New features may be added based on analytics or feedback. Thus, maintenance is **continuous and cyclical**, not a one-time task.

7. Describe the process of “Information Gathering.” Why is understanding the target audience crucial during this step?

Information gathering involves collecting requirements through client interviews, surveys, and market research. Understanding the **target audience** (age, tech literacy, needs, preferences) ensures the website’s design, content, and functionality align with user expectations—leading to better engagement and usability.

Group C: Scenario-Based Questions (5 Marks Each)

8. You are the project manager for a new e-commerce site. The client wants to skip the “Testing” phase to save money and launch early. Write a persuasive argument explaining the potential risks (security, functionality, compatibility) of skipping this phase.

Skipping testing risks:

- **Security flaws:** Unpatched vulnerabilities could lead to data breaches (e.g., stolen credit cards).
 - **Functionality bugs:** Broken checkout, cart errors, or payment failures frustrate users and lose sales.
 - **Compatibility issues:** Site may not work on mobile devices or certain browsers, excluding large user segments.
- Fixing these post-launch is **more expensive and damages brand reputation**. Testing ensures reliability, trust, and smooth user experience—critical for e-commerce success.

9. A developer begins coding a website immediately after meeting the client, without creating a design prototype or sitemap. Midway through, the client rejects the layout. Analyze what went wrong in the process and how it should have been handled.

The developer skipped essential **design and planning phases**. Without a wireframe or sitemap, there was no visual agreement on structure or layout. The correct approach: after initial discussion, create mockups/sitemaps for client approval **before coding**. This saves time, reduces rework, and aligns expectations early.

10. A university website was launched successfully, but after six months, students stopped visiting it because the exam schedules were outdated. Which phase of the development lifecycle was neglected? Explain the activities involved in this phase.

The **Maintenance phase** was neglected. Activities include:

- Regular content updates (e.g., exam dates, news)
 - Monitoring site performance and user feedback
 - Fixing broken links or outdated info
- Without scheduled maintenance, even a well-built site becomes irrelevant and loses user trust.
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CHAPTER 3: WEBSITE STRUCTURE DESIGN

Group A: Short Questions (2 Marks Each)

1. Define Information Architecture (IA).

Information Architecture (IA) is the practice of organizing, structuring, and labeling website content to support usability and findability—helping users locate information efficiently.

2. What is a “Wireframe”?

A wireframe is a low-fidelity sketch or blueprint of a webpage showing layout, content blocks, and navigation—without colors, images, or styling. It focuses on structure and user flow.

3. Explain the concept of “Cognitive Friction” in web design.

Cognitive friction occurs when users face confusion, delays, or mental effort due to poor design (e.g., unclear buttons, hidden menus). It increases bounce rates and reduces satisfaction.

4. Why should URL slugs use hyphens instead of underscores?

Search engines like Google treat **hyphens** (-) as word separators but interpret **underscores** (_) as part of a single word. So web-design is read as “web design,” while web_design may be seen as “webdesign”—hurting SEO and readability.

Group B: Long Questions (4 Marks Each)

5. Compare and contrast Hierarchical (Tree) structure and Linear (Sequential) structure. Give an example of when to use each.

- **Hierarchical:** Organized like a tree with a home page at the top and branches (categories → subpages). Ideal for large sites (e.g., university websites, e-commerce stores).
 - **Linear:** Pages follow a strict sequence (Page 1 → Page 2 → ...). Best for tutorials, storytelling, or step-by-step processes (e.g., online courses, product demos).
- Contrast:** Hierarchical allows free navigation; linear enforces order.

6. Explain the “Three-Click Rule” and its significance in User Experience (UX) design.

The Three-Click Rule suggests users should find any information within **three mouse clicks**

from the homepage. While not absolute, it emphasizes intuitive navigation—reducing frustration and abandonment. Good IA, clear menus, and search features help achieve this.

7. “Plan before you do.” Explain how tools like Card Sorting and Flowcharts help in planning a website’s structure.

- **Card sorting:** Users group content topics into categories, revealing natural mental models for organizing the site.
 - **Flowcharts:** Visualize user paths and page relationships, identifying redundant steps or missing links.
- Both tools ensure the final structure matches user expectations and supports efficient navigation.
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Group C: Scenario-Based Questions (5 Marks Each)

8. You are designing a website for a University. It has hundreds of pages (Admissions, Departments, Alumni, News). Which structural model would you choose? Draw a rough diagram and justify your choice.

Use a **Hierarchical (Tree)** structure.

Rough Diagram (text):

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9
- 10
- 11

Justification: A university has diverse, categorized content. A tree structure allows logical grouping, easy navigation, and scalability—users can drill down from broad to specific topics.

9. A user visits a website but leaves within 10 seconds because they cannot find the navigation menu, which is hidden behind a small icon on a desktop screen. Analyze this design failure using the “KISS” (Keep It Simple) principle.

The design violates **KISS** by overcomplicating navigation. On desktops, hiding the menu behind a hamburger icon (common on mobile) creates **cognitive friction**—users expect visible menus. Simplicity means making core functions **immediately obvious**, especially on larger screens where space isn’t limited.

10. An e-commerce website has a URL structure like www.shop.com/prod?id=55&cat=9. Explain why this is bad for both users and Search Engines (SEO), and propose a better structure using Page Slugs.

Problems:

- Not human-readable or memorable
- Lacks keywords for SEO
- Appears untrustworthy

Better structure:

www.shop.com/electronics/wireless-headphones

This uses descriptive **slugs**, improves SEO with keywords, enhances usability, and reflects site hierarchy.